Abbilash Shivaparan

Princeton, NJ • abbi.shiva@gmail.com • linkedin.com/in/abbi-shiva • github.com/abbishiva-oss • abbishiva-oss.github.io

Information Science student at the University of Maryland pursuing a minor in Data Science, with analytics experience spanning SQL, Excel reporting, and Power BI dashboarding, and Python-based automation for data cleaning and processing. Developed full-stack web applications and machine learning projects, and gained applied experience with retrieval-augmented generation (RAG) systems, large language models, and natural language processing through an industry AI externship and undergraduate research on LLM evaluation. Effective communicator with a track record of presenting data-driven insights to stakeholders.

EXPERIENCE

NJ TRANSIT • Data Analytics – IT Intern • 05/2024 – 09/2025

- Built SQL queries and Excel reports generating weekly operational statistics for 5,000+ ride records; automated geospatial validation of 2,000+ addresses using Haversine calculations, reducing manual review by 60%.
- Developed Python scripts using Google Reverse Geocoding API to rank top 25 high-frequency pickup locations for route optimization.
- Built 3 Power BI dashboards visualizing ridership trends and operational KPIs; created Crystal Reports dashboard flagging 150+ mismatched location IDs and presented enhancement roadmap to management.

Pfizer (via Extern) • AI-Powered Document Insights Extern • 06/2026 – 08/2026

- Selected for 10-week externship building AI document-analysis skills across RAG (retrieval-augmented generation) systems, large language models, NLP, and prompt engineering.
- Executed hands-on projects in document analysis and LLM integration, applying enterprise AI techniques to pharmaceutical knowledge-management use cases.
- Built a RAG pipeline over pharmaceutical certificate documents; resolved corrupted PDF text extraction by switching to PyMuPDF and increased retrieval depth from 3 to 10 chunks to recover multi-section answers.

Windsor Internal Medicine and Geriatrics • Office Assistant – IT Support • 06/2022 – Present

- Implemented access controls and security protocols for patient data protection; performed Windows patching across 10+ workstations maintaining HIPAA compliance.
- Resolved OS compatibility issues on corporate PCs, minimizing clinical staff downtime.

RESEARCH

University of Maryland, Chan Lab • Undergraduate Researcher (Advisor: Prof. Joel Chan) • 09/2026 – Present

- Evaluating how accurately frontier and local LLMs extract claims and evidence from research papers compared to human review, using 205 model-drafted discourse graph nodes across 27 papers.
- Leading local model setup (Gemma/Llama) and scoring with precision, recall, and F1; categorizing model errors by type, such as missing context, reframed meaning, and quote mismatches.
- Presented two progress talks and revised the study design after early review showed most AI-drafted nodes needed human edits.

PROJECTS

DataProcessBot – RAG-powered desktop app for document Q&A (Tauri, Python FastAPI, ChromaDB)

- Built a cross-platform desktop application enabling users to query their own documents through a retrieval-augmented generation pipeline powered by ChromaDB.

- Architected a Python FastAPI backend paired with a Tauri front end, delivering a lightweight, native-feeling desktop experience for document search and Q&A.
- Implemented grounded Q&A with relevance-threshold retrieval and source citations, plus auto-summarize, extract, and flag actions that run automatically on every upload.

SubleaseHer – Full-stack college sublease marketplace (Next.js 15, PostgreSQL, Prisma, NextAuth.js, Tailwind CSS) | Live: subleaseher.vercel.app

- Designed and deployed marketplace connecting students with housing; engineered PostgreSQL schema with Prisma ORM, integrated Cloudinary for image delivery, and implemented NextAuth.js authentication.
- Managed deployment via Vercel; maintained codebase on GitHub (abbishiva-oss).
- Built listing search with keyword, location, price, bed, and date filters, listing-based messaging with unread counts, favorites, and 1–5 star user reviews.

NBA Player DNA – ML clustering dashboard (Python, scikit-learn, Streamlit) segmenting 500+ players into 8 archetypes | Live: nba-player-dna.streamlit.app

- Implemented unsupervised pipeline: PCA for dimensionality reduction + K-Means clustering of player statistics; built interactive Streamlit dashboard with League Overview and Season Trends pages.
- Updated with 2025-26 season data; actively maintains bug fixes and feature additions.
- Built a Team Fit Analyzer scoring free-agent fit (55% archetype gap, 25% PIE, 10% points, 10% assists) within a 5-page dashboard that includes player comparison and shot charts.

Electric Vehicle Registration Analysis – SQL, database normalization (3NF), entity-relationship diagrams

- Designed normalized relational database for 100K+ WA State EV records; wrote complex SQL queries identifying adoption trends by county and vehicle type.
- Found top 3 manufacturers account for 78% of registrations; identified highest-adoption counties for planning insights.

PDF Research Buddy – Document Q&A chatbot (Python, Gradio, Gemini API, PyMuPDF)

- Extended a tutorial chatbot with multi-file upload and document switching, chat transcript export, a loading indicator, and a two-column Gradio Blocks interface.

EDUCATION

Bachelor of Science in Information Science, Minor in Data Science
University of Maryland • College Park, MD • Expected May 2027

- Activities: Cyber Security Club, AWS Practitioner Club, UMD Club Basketball

Relevant Coursework: Research in Information Science (INST408G), Database Design (INST327), Statistics (INST314), Data Science (DATA350), Data Ethics (DATA200), Information Organization (INST311)

SKILLS & CERTIFICATIONS

Data & BI: SQL, Excel (PivotTables, XLOOKUP), Power BI, Crystal Reports

Programming: Python, R, Java, JavaScript, React, Next.js, TypeScript

Databases & Web: PostgreSQL, Prisma ORM, NextAuth.js, Tailwind CSS, Cloudinary, FastAPI, Tauri, Zod

ML & Data Science: scikit-learn, pandas, PCA, K-Means, statistical analysis, Streamlit, RAG, LLMs, LlamaIndex, ChromaDB, embeddings, NLP

Tools: Git, GitHub, Oracle SQL Developer, Figma, Google APIs, Obsidian, Vercel

Certificates: Career Essentials in Data Analysis (Microsoft), Data Analytics Specialist (LinkedIn Learning)